AMT: Adaptive Multi-Sample Testing for Data Efficiency on Binomial Data

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Abstract. Multi-sample testing is a fundamental task in statistics, aiming to assess whether several samples taken under different conditions originate from a shared distribution. Classical methods for this task typically assume static, non-adaptive sampling and independent and identically distributed (iid) data. However, in many real-world applications (including materials science, medicine, and biology) data collection is costly, time-consuming, and often subject to experimental constraints, rendering static sampling inefficient. To address these limitations, we introduce Adaptive Multi-Sample Testing (AMT), a novel method that leverages Bayesian upper confidence bounds to dynamically allocate measurement effort to the most informative sample conditions. This adaptive approach induces statistical dependencies among samples, thereby violating the iid assumption underlying traditional tests. We overcome this challenge by developing a new test statistic that offers rigorous control of the type I error under adaptive, non-iid sampling schemes. Empirical results demonstrate that AMT achieves higher test power with fewer samples under the alternative hypothesis, while reliably maintaining type I error control under the null hypothesis. Overall, our approach provides a data-efficient solution for multi-sample testing in resource-constrained settings, with broad applicability across scientific domains.

Keywords: Hypothesis Testing · Adaptive Sampling

1 Introduction

In many scientific fields, such as biology [e.g., microarray studies; 16], medicine [e.g., clinical trials; 11] , and materials science [e.g., superconductivity classification; 24], a common task is that of multi-sample testing (MT) on binomial data, i.e., to determine whether multiple samples taken under different conditions originate from a shared distribution. However, a major challenge is that data collection is often costly and time-consuming, making it impractical to acquire the large sample sizes typically required for classic statistical methods. In materials science, for example, expensive materials are frequently lost during destructive measurements. Similarly, in medicine and psychology, conducting studies on humans is time-consuming and expensive. In addition, non-adaptive classic sampling methods, such as random or space-filling sampling, collect all data upfront. This often results in valuable resources being wasted on conditional samples that offer little information to distinguish between distributions, in turn reducing test power.

A promising direction for improving data efficiency is adaptive sampling, which, for example, [9] explores. Adaptive sampling operates iteratively through a feedback loop. First, it analyzes already collected data, then it decides where in the data space to collect additional data. However, its application to MT remains largely unexplored, which can be attributed to three primary challenges:
(P1) *Transferring adaptive sampling strategies to MT is not straightforward, because MT requires the non-trivial decision of which sample to add additional (new) data to, which differs from existing adaptive sampling tasks.*
(P2) *Because adaptive sampling introduces statistical dependence between observations, it violates the iid assumption required for many classic test statistics.*
(P3) *Test statistic agnostic permutation tests cannot be employed in their existing form since they also require iid data, or involve computationally demanding re-simulation of the entire adaptive sampling loop – see Sup.Mat. B.5.*

Our contributions to overcome these challenges are:

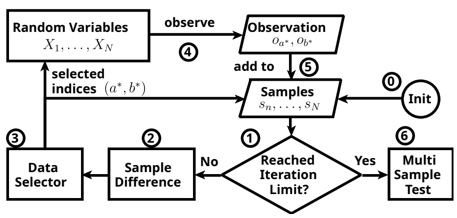


Fig. 1. Flow chart of our AMT approach.

(C1) We propose a novel Adaptive Multi-Sample Testing (AMT) approach, summarized in Figure 1. (0) AMT begins with N initial samples containing J observations each. (1) AMT checks a stopping criterion, e.g., a given budget of iterations. (2) If the criterion is not met, AMT estimates the difference between the variables based on the samples, based on a Bayesian upper confidence bound, that is, the posterior probability that a sample differs from the other samples. (3) AMT selects the two variables a^*, b^* with the highest difference, maximizing this probability, (4) observes the outcomes, and (5) adds them to the corresponding samples. (6) Once the iteration limit is met, ATM computes an MT using all N samples; thereby, AMT significantly reduces the amount of data required to achieve similar test power.
(C2) To obtain a test with theoretical guarantees for the type I error, while being computationally feasible, we need to correct for the non-iid samples generated by adaptive sampling. To achieve this, we derive the null distribution of each sample, conditioned on the sample size. We use this for sample-wise testing, combining the subtest results to form a Bonferroni-corrected omnibus test.

We emphasize that C1 can be considered a "bridging fields" contribution, adapting techniques from adaptive sampling and bandit algorithms to the sample selection problem in hypothesis testing. C2, on the other hand (step 6 in Figure 1), is our main contribution to the field of hypothesis testing, and, to our knowledge, a complete novelty. Existing (bandit) algorithms [1, 12] are not able to provide statistics or guarantees required for the type I error of a subsequent hypothesis test, and guarantees of existing test statistics [4, 13, 17] break the moment one acquires a sample with adaptive sampling.

As our title suggests, our work focuses on Bernoulli-distributed data, which is common in many MT applications. For example, in biology, one might use this to efficiently test whether multiple protein samples exhibit the same distribution of known protein sequences. However, AMT is conceptually also applicable in settings with observations from metric variables. This can be achieved by transforming the data into discrete "pseudo"-samples of binary observations, similarly to Mood's-Median test [15]. Details are given in Sup.Mat. B.2 and the method is evaluated on a real-life data set in a preliminary study.

Finally, we benchmark AMT in extensive numerical experiments against a range of adaptive and non-adaptive sampling strategies and test statistics. To demonstrate the flexibility and robustness of AMT, we also evaluate permutation-based variants of AMT. Our code and Sup.Mat. is available online[1].

2 Related Work

In this section, we review related work on MT, adaptivity in hypothesis testing, and discuss data-efficient decision-making techniques based on adaptive sampling through feedback loops from other disciplines.

Multi-Sample Testing: *Classic Approaches* include ANOVA [4], the Kruskal-Wallis test [a rank based nonparametric alternative; 13], the multi-sample chi-square test [17], and the median test [based on a chi-square test after data transformation; 15]. All these MT assume iid data requiring a non-adaptive sampling process. *Kernel-based and distribution-free tests* are applicable to higher dimensions, usually also requiring iid data. Examples are tests based on the maximum mean discrepancy [8] or its known equivalent, the energy distance [23]. *Permutation tests* are a general class of non-parametric tests, which generate a null distribution for virtually any chosen test statistic by randomly permuting the observations [2]. This idea relies on the exchangeability of observations under H_0, which is explicitly broken by adaptive sampling. Therefore, attaining a correct distribution under H_0 requires a non-standard permutation involving computationally intensive re-simulation of the adaptive sampling loop.

Adaptivity in Hypothesis Testing: Unlike classic approaches that use a fixed sample size, sequential designs [the origin of adaptive testing; 19] adapt to the data as it is collected to improve efficiency. More recent adaptive strategies, such as α-investing [5, 10], have focused on allocating an error budget across a series of multiple individual tests rather than MT, which involves a single global hypothesis test. Similar recent work on e-values [18], while useful for combining multiple individual tests, is not focused on sequential testing.

Adaptive sampling has shown promise in various areas, but adapting it to MT has proven challenging as it disrupts classic type I error control. Thus, most works focus on multi-sample detection, i.e., essentially MT but without type I

[1] Git: https://anonymous.4open.science/r/AMT_Adaptive-Multi-Sample-Testing-836D

error control. [9] is such a work using iterative sampling to detect differences between Gaussian variables, efficiently reducing the required sample size for detection, but missing the required type I error control for hypothesis testing.

Adaptive Sampling Techniques: Adaptive sampling aims to obtain (e.g., measure, label) data useful for data-efficiently solving a knowledge discovery task (e.g., model learning, optimum search). It has been used in the following contexts: *Multi-Armed Bandit Algorithms* sequentially select among multiple options (arms) to maximize a cumulative reward [14]. Some works address best-arm identification [1, 12], which is related to localizing a variable with stochastically different behavior. However, existing theoretical bounds are not tight enough for hypothesis testing. *Surrogate Model-Based Optimization* [7] and more specifically Bayesian optimization [22] aims to optimize expensive black-box functions with minimal data, rather than to perform statistical testing by adaptively exploring the input space. *Active Learning* shares the principle of adaptive data selection [21]. However its core focus is on prediction tasks, not hypothesis testing, seeking to train accurate predictive models with minimal labeled data.

In contrast to these tasks, we propose AMT, which utilizes adaptive sampling for MT, aiming to obtain data that aids the test decision while minimizing the sample size required for acceptable power. This requires a new and aligned understanding of the task-dependent concept of "data usefulness".

Summary and Contributions: To the best of our knowledge, no existing methods perform MT via adaptive data collection. The closest works using adaptive sampling [9] [12] focus on a form of multi-sample detection and do not provide type I error control in the context of hypothesis testing. In contrast, our work bridges the gap between adaptive methods and MT, providing a new test statistic with an analytically derived null distribution. Thereby, we directly address the lack of strict type I error control in an adaptive setting, with increased test power, and offer a robust solution for MT in data-limited scenarios.

3 Preliminaries

In this section, we provide the basics of Bernoulli Multi-Sample Testing.

Multi-Sample Testing (MT), as the name suggests, requires multiple given samples s_n, $n \in \{1, \ldots, N\}$ of size $|s_n|$. MT assumes that a sample is a set of observations $s_n = \{o_1, o_2, \ldots, o_{|s_n|}\}$ from a random variable X_n. Obtaining a single observation o from the random variable X_n is denoted as $o \leftarrow X_n$, while obtaining a whole sample s_n of size J is denoted as $s_n \underset{J}{\leftarrow} X_n$. MT generally has the goal of determining whether all samples (and thus all random variables) originate from a shared distribution $X_1, \ldots, X_N \sim P(x|\theta)$ with parameter $\theta \in \mathbf{\Theta}$ or not. MT does not aim to identify which specific samples differ, but rather to detect whether any deviation exists. Post-hoc analysis may be used to localize differences once H_0 is rejected. Sup.Mat. B.1 gives some guidelines on MT.

In practice, such as in biology [16], medicine [11] , and materials science [24], one is often faced with binary observations $o \in \{0, 1\}$ and Bernoulli random variables $X_n \sim \text{Bernoulli}(p_n)$, with success probability p_n, leading to binomial distributed samples $s_n \underset{J}{\leftarrow} X_n$. Here, θ then refers to the success probability p_n or sample mean $\bar{p}_n$, and one is interested in testing whether the sample means differ. More formally, this leads to the following definition:

Definition 1 (Bernoulli Multi-Sample Testing (BMT)). *Let $p_1, \ldots, p_N$ be the true success probabilities of N independent Bernoulli-distributed random variables $X_1, \ldots, X_N$. The goal is to test*

$$H_0 : \ p_1 = p_2 = \cdots = p_N = p \quad \textit{vs.} \quad H_1 : \ \exists\, n \in \{1, \ldots, N\}, \ \textit{such that } p_n \neq p,$$

$$\textit{where } p = \frac{1}{N} \sum_{n=1}^{N} p_n \textit{ is the mean of } p_1, \ldots, p_N.$$

Remark 1. BMT can be interpreted as Fake Coin Detection (FCD), where we think of each random variable X_n as a coin that can be tossed, and its outcome observed (i.e., head or tail). BMT then tests whether all coins are the same or if one coin is plated. In Sup.Mat. B.2, we describe how one can use BMT on metric variables using a transformation similar to Mood's-Median test [15].

4 Our Method: Adaptive Multi-Sample Testing (AMT)

This section gives a detailed overview of AMT and the building blocks of the resulting Algorithm 1, including sample initialization, adaptive coin selection, the coin difference metric, and our test statistic used in the final multi-sample test, which corrects for adaptive sampling. This section will then focus on our new test statistic, details on our new adaptive sampling approach are given in Sup.Mat. B.3. We close the section with two remarks, one on the relevance and constraints of AMT, and the other on bounds for the detectable coin difference.

The steps of AMT in Algorithm 1 are as follows:
(1) Init: Let N be the number of Bernoulli coins X_n with $n \in \{1, \ldots, N\}$, I be the maximum number of sampling iterations, and J be the number of initial observations per sample s_n such that $|s_n| = J$ after initialization (Line 1). In Lines 2 to 3 each sample s_n is initialized with J observations o_a with $a \in \{1, \ldots, J\}$ from coin X_n: $s_n \underset{J}{\leftarrow} X_n$. For a sample s_n we denote by $h_n = \sum_{o_a \in s_n} o_a$ the number of heads and by $t_n = |s_n| - h_n$ the number of tails.
(2) Stopping Criterion: At the beginning of each iteration (Line 4) AMT checks if the given iteration limit I has been reached, and will continue sampling if not. Here, I is a proxy for the problem dependent maximum combined sample size $|S| = \sum_{n=1}^{N} |s_n|$, which can be determined through a power analysis.
(3) Coin Selection: In Line 5, AMT selects two coins X_{a^*} and X_{b^*} that maximize a predefined *coin difference* $d(s_a, s_b)$ between their current respective samples s_a and s_b. This selection is formally expressed as: $a^*, b^* = \arg\max_{a,b} d(s_a, s_b)$. We construct $d(s_a, s_b)$ such that it estimates the probability of two coins having different success probabilities, making it a variant of Bayesian confidence bounds; see Sup.Mat. B.3 for details. Sampling according to these bounds automatically balances exploration and exploitation, eliminating the need for manual tuning.

Algorithm 1 AMT: Adaptive Multi-Sample Testing

1: **procedure** AMT(N, I, J) ▷ *Inputs: N, I, J*
2: **for** $n \in \{1, \ldots, N\}$ **do**
3: $s_n \overset{J}{\leftarrow} X_n$ ▷ *Sample Initialization*
4: **for** $i \in \{1, \ldots, I\}$ **do** ▷ *Stopping Criterion*
5: $(a^*, b^*) := \arg\max_{a,b} d(s_a, s_b)$ ▷ *Coin Selection*
6: $o_{a^*} \leftarrow X_{a^*}$
7: $o_{b^*} \leftarrow X_{b^*}$ ▷ *Coin Sampling*
8: $s_{a^*} := s_{a^*} \cup \{o_{a^*}\}$
9: $s_{b^*} := s_{b^*} \cup \{o_{b^*}\}$ ▷ *Sample Update*
10: MSTest($s_1, s_2, ..., s_N$) ▷ *Multi-Sample Test*

We note that our algorithm is flexible in the sense that it can utilize different choices for the difference $d(s_a, s_b)$ from other fields, resulting in different variants of AMT. We explore such variants in our experiments in Section 5.

(4) Coin Sampling: Subsequently, AMT observes the two selected coins X_{a^*} and X_{b^*} (Lines 6 to 9), adding the outcomes o_{a^*} and o_{b^*} to their respective sample sets $s_{a^*} := s_{a^*} \cup \{o_{a^*}\}$ and $s_{b^*} := s_{b^*} \cup \{o_{b^*}\}$.

(5) Multi-Sample Test: Upon reaching the iteration limit, a multi-sample test is performed on the collected samples (Line 10). Although AMT can use various existing test statistics (as demonstrated in Section 5.1), adaptive sampling results in non-iid data, which invalidates the null distribution of classic tests. To correct for this, we propose a novel test statistic, based on the Beta-Binomial distribution, that maintains comparable power while allowing for the analytical derivation of the null distribution and, thereby, the critical values for controlled type I error. We give the details in Section 4.1. Compared to permutation testing, our proposal is much faster and has less computational complexity, as exemplified in Sup.Mat. B.5.

Remark 2 (Relevance and Constraints). AMT shares the same goals as MT, with the added objective of arriving at a meaningful test decision with as little data as possible. However, by doing so, AMT is constrained to applications where additional data can be collected on request (in contrast to only having observational data). This constraint is often fulfilled in experimental science, for which AMT is also most relevant, because here data collection is often expensive.

Remark 3 (Detectable Coin Difference). A result from [9] suggests that adaptive sampling is able to detect smaller differences $\delta p = |p_a - p_b|$ between coins than classic non-adaptive methods, which are theoretically limited by the combined sample size $|S|$. Sup.Mat. B.4 outlines how to adapt this result to AMT.

4.1 Multi-Sample Test Statistic

A closed-form solution for the distribution under H_0 proved to be intractable. Hence, we first perform separate single coin tests for each $n = 1, \ldots, N$. Formally, these tests are defined as follows:

Definition 2 (Single Coin Test). *Let p_n be the success probability of coin X_n, and $p_{\neg n} = \frac{1}{N-1} \sum_{k \in \{1,\ldots,N\}, k \neq n} p_k$. A single coin test considers*

$$H_{0n}: \; p_n = p_{\neg n} \quad vs. \quad H_{1n}: \; p_n \neq p_{\neg n}.$$

Here, H_{0n} is derived from H_0, where $p_n = p_{\neg n} = p$. Aggregating the single coin test results allows us in the second step to construct an omnibus test for H_0, where we reject H_0 if and only if any of the H_{0n} were rejected. To control the overall significance level under H_0, we apply the Bonferroni correction, setting $\alpha_n = \alpha/N$ for each single coin test [6]. As the Bonferroni correction is known to be conservative, we refer to the resulting reduction in power in our experiments and investigate it more closely in Sup.Mat. D.5.

The following theorem specifies the distribution of h_n under H_{0n} (and H_0), under adaptive sampling, given that the shared p is known.

Theorem 1 (Distribution of h_n under H_{0n}). *Let $H_{0n} : p = p_n$ be the null hypothesis, where p is given. Assume that the sampling strategy depends only on previous observations and not on the outcome of the current trial. Assume further that at iteration I the number of trials per coin $|s_n|$ is known. Then, for a coin X_n, the number of heads h_n in the sample s_n after $|s_n|$ trials follows a Binomial distribution: $h_n \sim \mathrm{Bin}(|s_n|, p)$.*

The proof of Theorem 1 is given in Sup.Mat. A.

In practice, the true parameter p is unknown, requiring a posterior distribution $\mathrm{Beta}(a_{\neg n}, b_{\neg n})$ of p. First, we build the combined sample set $s_{\neg n} = \bigcup_{k \in \{1,\ldots,N\}, k \neq n} s_k$, which explicitly excludes s_n, and compute the number of heads $h_{\neg n} = \sum_{o_k \in s_{\neg n}} o_k$ and the number of tails $t_{\neg n} = |s_{\neg n}| - h_{\neg n}$ in $s_{\neg n}$. Assigning a Beta prior $\mathrm{Beta}(a, b)$ results in the posterior distributions $\mathrm{Beta}(a_{\neg n}, b_{\neg n})$ of $p \mid s_{\neg n}$ with parameters $a_{\neg n} = 1 + h_{\neg n}, b_{\neg n} = 1 + t_{\neg n}$. Combining these Beta posteriors with the Binomial distributions under H_{0n} of Theorem 1 yields a Beta-Binomial posterior distribution for h_n:

$$h_n \sim \mathrm{BetaBin}(|s_n|, a_{\neg n}, b_{\neg n}).$$

We can use $\mathrm{BetaBin}(|s_n|, a_{\neg n}, b_{\neg n})$ to determine the upper and lower critical values at $\alpha_n/2$-levels for each single coin test, compare them against the observed number of heads h_n, and finally combine the results into our omnibus test.

5 Experiments

The following section describes the baselines and variants, experimental settings and evaluation metrics, and the main results.

5.1 Baselines and Variants

By replacing steps (3)–(5) in our Algorithm 1, one can mimic different baselines [corresponding to existing methods] and create variants of AMT [different versions of AMT]. For brevity, we will only give a overview of relevant methods; Sup.Mat. C.1 gives details and additional used baselines.

As test statistics we use: *(1) the χ^2 statistic*, *(2) the KW statistic*, *(3) the difference in means (Mean)* between each coin and the coin average, and *(4)* *the Beta-dist statistic* which is the "counterpart" to our Beta sampling based on distributional distance, see Sup.Mat. C.1. We compare them with our Beta-Binomial statistic. For all baselines and variants, we use a permutation testing approach based on 10000 re-simulations of the adaptive sampling process to obtain a good approximation under H_0 and to guarantee the required type I error. Our new test statistic, with its H_{0n} distributions, is not reliant on such computationally expensive re-simulation.

For coin selection, we use the classic non-adaptive sampling strategy *Space Filling **(Equal)***, and standard methods for adaptive sampling: *(1) Greedy*, and *(2) Vanishing Epsilon Greedy **(Slow)***. We combine them with different coin differences: *(1) Difference in Mean*, and *(2) our proposed posterior distance **(Beta)*** from Sup.Mat. B.3. Furthermore, we apply *(3) Thompson sampling **(TS)*** with Beta prior [20], which is a standard method in the Bandit setting, and *(4) TS with five times higher exploitation **(TS5)***.

5.2 Experimental Settings and Evaluation Metrics

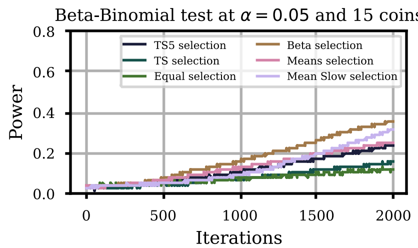
All experiments were run using Python under Linux on an 8-core AMD Ryzen 7 4750U and 15 GB of RAM. We vary the number of coins ($N \in [5, 10, 15, 20]$), the number of iterations before the Multi-Sample Test is performed ($I \in (1 \dots 2000)$), the number of biased coins ($M \in [0, 1, 2, N/2]$), the difference in probability ($\delta p \in [0.15, 0.1, 0.05]$), and the significance level ($\alpha \in [0.05, 0.025, 0.01]$). ATM was additionally evaluated on the real-life Coffee Data [3].

To calculate the power and type I error, we perform each setup 10000 times under H_0 and H_1. We evaluate the test statistic and coin selection per iteration. With this, we plot the empirical distribution under H_0, H_1, and calculate the critical values, power, type I error, and distance between H_0, H_1. Details are given in Sup.Mat. C.2. We investigate changes in distributions under H_0 due to adaptive sampling. We compare the results between different variants and non-adaptive baselines. For each variant V, we report the power gained against the non-adaptive Equal sampling: $\text{gain}_V = (\text{power}_V - \text{power}_{Eq})/\text{power}_{Eq}$.

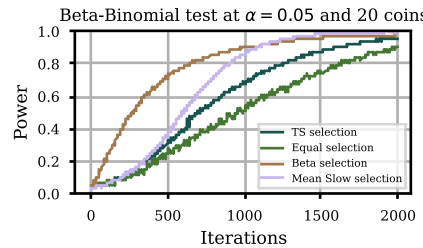
5.3 Results

The results presented here will focus on the power of the tests. Further results and ablation studies are provided in Sup.Mat. D. These include verification of the controlled type I error and analysis of the impact of adaptive sampling on the H_0 and H_1 distributions.

Table 1 (page 11) provides a large-scale overview of all results, by reporting the power of adaptive sampling to non-adaptive sampling for different test statistics and sampling methods. The table shows how much power we gain by using adaptive sampling at an type I error rate of $\alpha = 5\%$ and a difference between coins of $\delta p = 0.05$, with one plated coin ($M = 1$). We vary the number of coins N, and the number of iterations I, at which the test is performed. We see that in any scenario, every adaptive strategy is at least as powerful as the corresponding non-adaptive strategy, with a gain of up to 485%. In particular, our Beta sampling outperforms all other strategies (marked in bold) across all test statistics. Finally, we observe that the gain increases with the number of iterations after which the test is performed. This observation is expected as sampling strategies transition from exploration to exploitation.



(a) $\delta p = 0.05$



(b) $\delta p = 0.15$

Fig. 2. The impact of different sampling strategies.

Power behavior over sample size: Figure 2a investigates the power of our Beta-Binomial test with various sampling strategies for a range of sample sizes, with $m = 1$, $n = 15$, and $\delta p = 0.05$. The sampling strategy significantly impacts the resulting power: A direct comparison of TS and TS5 shows that the slightly more exploitative TS5 strategy provides a increase in power. Conversely, comparing the purely exploitative Mean strategy to the Mean Slow strategy (a linearly vanishing ϵ-greedy method) reveals that excessive exploitation harms power. This highlights the well-known exploration-exploitation tradeoff [25] and the sensitivity of these methods to hyperparameter selection. However, our Beta sampling does not require such a hyperparameter and consistently performs best. Similarly, in Figure 2b for $\delta p = 0.15$, we can observe the full power spectrum and see how our Beta sampling gains power much faster, only requiring *half* the sample size of Equal sampling to arrive at a power of 80%.

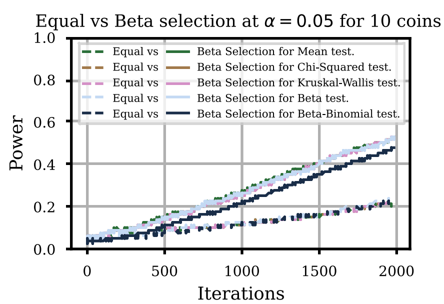


Fig. 3. Beta vs. Equal sampling, $\delta p = 0.05$.

Comparison of test statistics under both non-adaptive (Equal) and adaptive (Beta) sampling: From Figure 3, we can conclude that the choice of test statistic has a minimal impact on power when the sampling method is fixed, with only a slight deviation observed in our Beta-Binomial statistic. This suggests that the sampling strategy itself is the primary driver of power, not the specific test statistic. Under non-adaptive sampling, all test statistics result in similar power. Under adaptive sampling, our Beta-Binomial statistic is slightly less powerful than the other permutation-based statistics. This is somewhat expected because it builds on the Bonferroni correction, which is known to be conservative [6], and thus contributes to the reduction in power, as detailed in Sup.Mat. D.5. Nonetheless, it still significantly outperforms all statistics under non-adaptive sampling approach, offering the distinct advantage of being based on the analytic H_{0n} distributions, thus requiring far less computational power than permutation testing.

6 Conclusion

We propose Adaptive Multi-Sample Testing (AMT), a novel multi-sample testing approach that dynamically focuses measurement efforts on the most informative sample conditions. This adaptive process differs fundamentally from classic static sampling, thereby violating the iid assumption. We address this challenge with a new test statistic that provides theoretical guarantees for type I error control. Our approach is particularly valuable in domains where data acquisition is expensive, such as materials science and medicine. However, the adaptivity of AMT is at the same time its primary constraint, i.e., it requires access to a data source from which new additional data can be obtained on request. Inherently, this constraint is met in most experimental sciences targeted by AMT. Our empirical results demonstrate that AMT gains up to *485%* in test power, requiring *half* the data compared to classic approaches, confirming that adaptive sampling can significantly improve the power of statistical tests while maintaining type I error control. Looking forward, we believe expanding adaptive sampling to hypothesis tests other than MT is a promising research direction.

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Table 1. Power and Gain of Multi-Sample Tests with Adaptive vs Non-Adaptive Sampling at $\delta p = 0.05$

		Mean		**Chi-squared**		**Kruskal-Wallis**		**Beta-Dist**		**Beta-Binomial**	
N / I	**Method**	**Power**	**Gain (%)**	**Power**	**Gain (%)**	**Power**	**Gain (%)**	**Power**	**Gain (%)**	**Power**	**Gain (%)**
10 / 1000	Equal	0.068	–	0.068	–	0.068	–	0.071	–	0.073	–
	TS5	0.166	144	0.144	111	0.144	111	0.168	138	0.132	80
	TS	0.100	47	0.092	34	0.092	34	0.101	42	0.088	20
	Beta	**0.201**	**196**	**0.189**	**177**	**0.189**	**177**	**0.198**	**180**	**0.166**	**126**
	Means	0.187	175	0.174	155	0.174	155	0.173	145	0.137	87
	Mean Slow	0.102	51	0.100	46	0.100	46	0.113	60	0.097	33
15 / 1000	Equal	0.043	–	0.043	–	0.043	–	0.048	–	0.045	–
	TS5	0.105	146	0.081	88	0.081	88	0.102	114	0.080	76
	TS	0.050	17	0.047	10	0.047	10	0.056	17	0.045	0
	Beta	**0.154**	**262**	**0.139**	**224**	**0.139**	**224**	**0.145**	**205**	**0.121**	**167**
	Means	0.134	215	0.133	211	0.133	211	0.130	174	0.096	113
	Mean Slow	0.067	57	0.068	58	0.068	58	0.075	57	0.059	30
20 / 1000	Equal	0.041	–	0.041	–	0.041	–	0.029	–	0.027	–
	TS5	0.075	85	0.064	58	0.064	58	0.081	183	0.056	111
	TS	0.042	3	0.042	3	0.042	3	0.045	57	0.034	27
	Beta	**0.131**	**222**	**0.120**	**195**	**0.120**	**195**	**0.126**	**342**	**0.099**	**272**
	Means	0.119	194	0.110	171	0.110	171	0.110	285	0.077	189
	Mean Slow	0.063	56	0.057	40	0.057	40	0.066	131	0.053	100
10 / 1500	Equal	0.101	–	0.101	–	0.101	–	0.100	–	0.110	–
	TS5	0.244	142	0.230	129	0.230	129	0.261	161	0.238	116
	TS	0.146	45	0.136	35	0.136	35	0.148	48	0.144	32
	Beta	**0.326**	**223**	**0.320**	**218**	**0.320**	**218**	**0.337**	**237**	**0.286**	**160**
	Means	0.265	163	0.259	157	0.259	157	0.255	155	0.220	100
	Mean Slow	0.201	100	0.192	91	0.192	91	0.218	118	0.209	90
15 / 1500	Equal	0.051	–	0.052	–	0.052	–	0.063	–	0.062	–
	TS5	0.168	228	0.141	172	0.141	172	0.167	166	0.135	118
	TS	0.071	39	0.069	33	0.069	33	0.090	43	0.074	20
	Beta	**0.256**	**397**	**0.245**	**375**	**0.245**	**375**	**0.241**	**283**	**0.214**	**247**
	Means	0.203	295	0.204	296	0.204	296	0.201	220	0.160	159
	Mean Slow	0.145	181	0.134	159	0.134	159	0.165	162	0.144	133
20 / 1500	Equal	0.042	–	0.042	–	0.042	–	0.042	–	0.039	–
	TS5	0.115	175	0.099	138	0.099	138	0.129	208	0.095	142
	TS	0.058	40	0.056	35	0.056	35	0.059	41	0.050	27
	Beta	**0.206**	**396**	**0.182**	**338**	**0.182**	**338**	**0.191**	**354**	**0.160**	**306**
	Means	0.174	318	0.162	289	0.162	289	0.159	278	0.125	217
	Mean Slow	0.111	166	0.105	153	0.105	153	0.126	200	0.105	166
10 / 2000	Equal	0.139	–	0.139	–	0.139	–	0.162	–	0.151	–
	TS5	0.356	156	0.331	138	0.331	138	0.381	135	0.348	131
	TS	0.202	45	0.194	40	0.194	40	0.226	39	0.208	38
	Beta	**0.454**	**227**	**0.457**	**229**	**0.457**	**229**	**0.468**	**188**	**0.414**	**174**
	Means	0.347	150	0.348	150	0.348	150	0.346	113	0.302	100
	Mean Slow	0.373	168	0.368	165	0.368	165	0.391	141	0.369	144
15 / 2000	Equal	0.064	–	0.064	–	0.064	–	0.081	–	0.077	–
	TS5	0.220	245	0.200	214	0.200	214	0.239	195	0.201	161
	TS	0.106	67	0.100	56	0.100	56	0.131	61	0.108	39
	Beta	**0.339**	**432**	**0.329**	**416**	**0.329**	**416**	**0.337**	**316**	**0.306**	**297**
	Means	0.262	310	0.269	322	0.269	322	0.260	221	0.219	183
	Mean Slow	0.256	301	0.234	267	0.234	267	0.289	256	0.261	239
20 / 2000	Equal	0.049	–	0.049	–	0.049	–	0.057	–	0.048	–
	TS5	0.162	227	0.137	177	0.137	177	0.173	204	0.142	197
	TS	0.072	46	0.069	40	0.069	40	0.081	41	0.067	41
	Beta	**0.289**	**485**	**0.274**	**454**	**0.274**	**454**	**0.280**	**391**	**0.249**	**421**
	Means	0.219	343	0.215	335	0.215	335	0.213	274	0.177	271
	Mean Slow	0.213	331	0.193	290	0.193	290	0.231	306	0.195	309

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